

```

#LoRA- CPU (Tiny version)
# TinyLlama check
from transformers import AutoModelForCausalLM
model = AutoModelForCausalLM.from_pretrained(
    "TinyLlama/TinyLlama-1.1B-Chat-v1.0"
)

```

... /usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret 'HF_TOKEN' does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set it as secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.

warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
config.json: 100% ██████████ 608/608 [00:00<00:00, 31.5kB/s]

model.safetensors: 100% ██████████ 2.20G/2.20G [00:35<00:00, 60.9MB/s]

Loading weights: 100% ██████████ 201/201 [00:00<00:00, 329.39it/s, Materializing param=model.norm.weight]

generation_config.json: 100% ██████████ 124/124 [00:00<00:00, 12.5kB/s]

```

import transformers
print(transformers.__version__)

5.0.0

```

```

import torch
from datasets import load_dataset
from transformers import (
    AutoTokenizer,
    AutoModelForCausalLM,
    TrainingArguments,
    Trainer,
    DataCollatorForLanguageModeling,
)
from peft import LoraConfig, get_peft_model

```

```

import os
print(os.path.exists("train.jsonl"))

True

```

```

model_name = "TinyLlama/TinyLlama-1.1B-Chat-v1.0"
data_path = "train.jsonl"
output_dir = "./tinyllama_lora_cpu"

```

```

# For CPU
device = torch.device("cpu")
torch.set_num_threads(4)

```

```

model = AutoModelForCausalLM.from_pretrained(
    model_name,
    torch_dtype=torch.float32,
)
model.to(device)

```

'torch_dtype' is deprecated! Use 'dtype' instead!

Loading weights: 100% ██████████ 201/201 [00:09<00:00, 15.75it/s, Materializing param=model.norm.weight]

```

LlamaForCausalLM(
  (model): LlamaModel(
    (embed_tokens): Embedding(32000, 2048)
    (layers): ModuleList(
      (0-21): 22 x LlamaDecoderLayer(
        (self_attn): LlamaAttention(
          (q_proj): Linear(in_features=2048, out_features=2048, bias=False)
          (k_proj): Linear(in_features=2048, out_features=256, bias=False)
          (v_proj): Linear(in_features=2048, out_features=256, bias=False)
          (o_proj): Linear(in_features=2048, out_features=2048, bias=False)
        )
        (mlp): LlamaMLP(
          (gate_proj): Linear(in_features=2048, out_features=5632, bias=False)
          (up_proj): Linear(in_features=2048, out_features=5632, bias=False)
          (down_proj): Linear(in_features=5632, out_features=2048, bias=False)
          (act_fn): SiLUActivation()
        )
        (input_layernorm): LlamaRMSNorm((2048,), eps=1e-05)
        (post_attention_layernorm): LlamaRMSNorm((2048,), eps=1e-05)
      )
    )
    (norm): LlamaRMSNorm((2048,), eps=1e-05)
    (rotary_emb): LlamaRotaryEmbedding()
  )
  (lm_head): Linear(in_features=2048, out_features=32000, bias=False)
)

```

```

# LoRA config
lora_config = LoraConfig(
    r=4,
    lora_alpha=8,
    lora_dropout=0.05,
    bias="none",
    task_type="CAUSAL_LM",
    target_modules=["q_proj", "v_proj"],
)

```

Requirement already satisfied: torchao in /usr/local/lib/python3.12/dist-packages (0.17.0)

```

trainable params: 1,126,400, all params: 1,100,174,784 / trainable: 0.1023
/usr/local/lib/python3.12/dist-packages/peft/tuning_func.py:72: UserWarning: You are trying to modify a model with PEFT for a second time. If you want to reload the model with a different config, make sure to call ".unload()" before.
  warnings.warn()
/usr/local/lib/python3.12/dist-packages/peft/tuning_func.py:78: UserWarning: The PEFT config's 'base_model_name_or_path' was renamed from 'TinyLlama/TinyLlama-1.1B-Chat-v1.0' to 'None'. Please ensure that the correct base model is loaded.
  warnings.warn()
/usr/local/lib/python3.12/dist-packages/peft/tuners/tuners_utils.py:302: UserWarning: Already found a 'peft_config' attribute in the model. This will lead to having multiple adapters in the model. Make sure to know what you are doing.
  warnings.warn()

```

```
dataset = load_dataset("json", data_files=({"train": data_path}))["train"]
```

Step	Training Loss
1	0.0000
2	0.0000
3	0.0000
4	0.0000
5	0.0000
6	0.0000
7	0.0000
8	0.0000
9	0.0000
10	0.0000
11	0.0000
12	0.0000
13	0.0000
14	0.0000
15	0.0000
16	0.0000
17	0.0000
18	0.0000
19	0.0000
20	0.0000
21	0.0000
22	0.0000
23	0.0000
24	0.0000
25	0.0000
26	0.0000
27	0.0000
28	0.0000
29	0.0000
30	0.0000
31	0.0000
32	0.0000
33	0.0000
34	0.0000
35	0.0000
36	0.0000
37	0.0000
38	0.0000
39	0.0000
40	0.0000
41	0.0000
42	0.0000
43	0.0000
44	0.0000
45	0.0000
46	0.0000
47	0.0000
48	0.0000
49	0.0000
50	0.0000
51	0.0000
52	0.0000
53	0.0000
54	0.0000
55	0.0000
56	0.0000
57	0.0000
58	0.0000
59	0.0000
60	0.0000
61	0.0000
62	0.0000
63	0.0000
64	0.0000
65	0.0000
66	0.0000
67	0.0000
68	0.0000
69	0.0000
70	0.0000
71	0.0000
72	0.0000
73	0.0000
74	0.0000
75	0.0000
76	0.0000
77	0.0000
78	0.0000
79	0.0000
80	0.0000
81	0.0000
82	0.0000
83	0.0000
84	0.0000
85	0.0000
86	0.0000
87	0.0000
88	0.0000
89	0.0000
90	0.0000
91	0.0000
92	0.0000
93	0.0000
94	0.0000
95	0.0000
96	0.0000
97	0.0000
98	0.0000
99	0.0000
100	0.0000

1	3.416027
2	1.873944
3	2.053949
4	1.666748

학습 완료
저장 위치: ./tinyllama_lora_cpu

```
tokenized_dataset = dataset.map(
    tokenize_function,
    remove_columns=["text"]
)
```

Map: 100% 4/4 [00:00<00:00, 35.25 examples/s]

```
dataset = dataset.map(
    tokenize,
    remove_columns=dataset.column_names
)
```

Map: 100%  4/4 [00:00<00:00, 109.39 examples/s]

```
# Training argument
training_args = TrainingArguments(
    output_dir = output_dir1,
    per_device_train_batch_size=1,
    gradient_accumulation_steps=1,
    num_train_epochs=1,
    # □□□□
    learning_rate=5e-4,
    logging_steps=1,

    save_steps=20,
    save_total_limit=1,
    fp16=False,
    bf16=False,
    report_to="none",
)
```

```
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=dataset,
    data_collator=DataCollatorForLanguageModeling(
        tokenizer=tokenizer,
        mlm=False
    ),
)
```

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WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Generating train split: 4/0 [00:00<00:00, 35.23 examples/s]
Map: 100% [████████████████████████████████████████] 4/4 [00:00<00:00, 36.84 examples/s]
'torch_dtype' is deprecated! Use 'dtype' instead!
Loading weights: 100% [████████████████████████████████████████] 201/201 [00:11<00:00, 20.58it/s, Materializing param=model.norm.weight]
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true but no accelerator is found, then device pinned memory won't be used.
  super().__init__(loader)
[████████████████████████████████████████] [4/4 00:44, Epoch 1/1]

Step   Training Loss
-----
1      3.416027
2      1.875223
3      2.061383
4      1.670421

✓ 완료! 저장 위치: ./tinyllama_lora_output
```